

Credit Card Default Auto-ML Report — ensemble (sub)

 2026-05-28 22:10:00 UTC

모형 ensemble

roc_auc = 0.7663

BEST(catboost) = 0.7796

Δ vs best = -0.0133

1 점수 요약 (본 모형 vs best)

metric	ensemble (현재)		catboost (BEST)		Δ(Test)
	Train	Test	Train	Test	
roc_auc	0.7902	0.7663	0.8099	0.7796	-0.0133
ks	0.4389	0.4168	0.4660	0.4362	-0.0195
lift	3.2341	3.0595	3.4037	3.1274	-0.0678
accuracy	0.8248	0.8153	0.8308	0.8177	-0.0023
precision	0.7058	0.6585	0.7133	0.6589	-0.0005
recall	0.3569	0.3429	0.3931	0.3640	-0.0211
f1	0.4741	0.4509	0.5069	0.4689	-0.0180
pr_auc	0.5809	0.5377	0.6194	0.5588	-0.0211

2 Feature Importance — ensemble

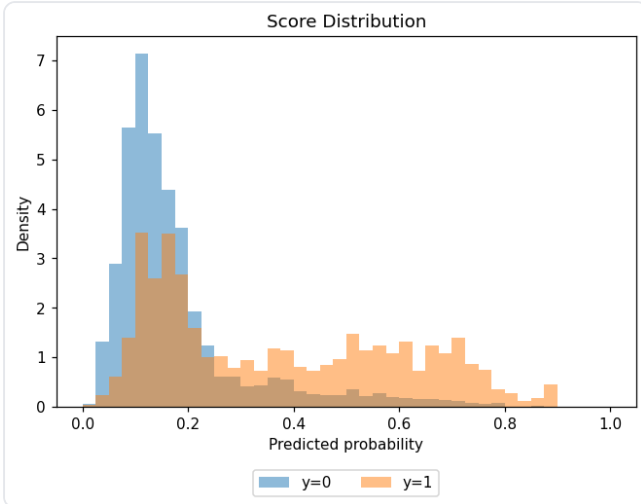
relative · top 19 · 전체 분포는 feature_importance.csv

Rank	Feature	Importance	Rank	Feature	Importance
1	PAY_0	0.2307	16	BILL_AMT5	0.0264
2	LIMIT_BAL	0.1036	17	BILL_AMT6	0.0260
3	BILL_AMT1	0.0899	18	PAY_AMT5	0.0224
4	PAY_2	0.0576	19	SEX	0.0218
5	PAY_3	0.0498			
6	PAY_AMT3	0.0473			
7	PAY_AMT2	0.0470			
8	PAY_AMT1	0.0411			
9	BILL_AMT2	0.0403			
10	PAY_4	0.0395			
11	AGE	0.0342			
12	PAY_AMT6	0.0326			
13	PAY_6	0.0314			
14	PAY_5	0.0300			
15	PAY_AMT4	0.0284			

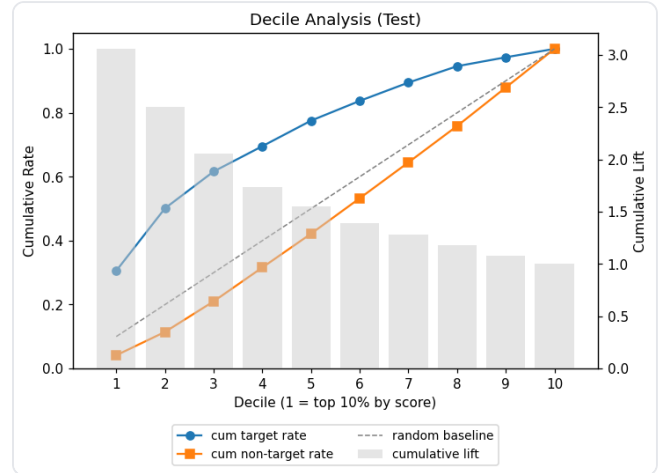
3 모델 상세 — ensemble

점수 분포 · 10-분위 차트

Score Distribution by Label



Decile Analysis Chart



Decile	Score Range	Count	Target	Non-Target	Cum Target	Cum NonTarget	Cum Count	Cum %
1	0.526 ~ 0.895	600	406	194	406	194	600	10.0%
2	0.328 ~ 0.526	600	259	341	665	535	1200	20.0%
3	0.213 ~ 0.328	600	153	447	818	982	1800	30.0%
4	0.177 ~ 0.213	600	105	495	923	1477	2400	40.0%
5	0.153 ~ 0.177	600	106	494	1029	1971	3000	50.0%
6	0.132 ~ 0.153	600	82	518	1111	2489	3600	60.0%
7	0.114 ~ 0.132	600	76	524	1187	3013	4200	70.0%
8	0.099 ~ 0.114	600	68	532	1255	3545	4800	80.0%
9	0.079 ~ 0.099	600	37	563	1292	4108	5400	90.0%
10	0.021 ~ 0.079	600	35	565	1327	4673	6000	100.0%

Confusion Matrix threshold = 0.5

	Pred 0	Pred 1
True 0	4437	236
True 1	872	455

4 변수 선택 (Stability Selection)

base: lasso

부분표본 80회

임계값 0.60

채택 19 / 23

표 노출 23 / 전체 23 (전체 분포는 메인 리포트의 feature_selection.csv)

Feature		Selection Frequency	채택
LIMIT_BAL		100.0%	✓ 채택
AGE		100.0%	✓ 채택
PAY_0		100.0%	✓ 채택
PAY_2		100.0%	✓ 채택
PAY_3		100.0%	✓ 채택
PAY_4		100.0%	✓ 채택
BILL_AMT1		100.0%	✓ 채택
PAY_AMT1		100.0%	✓ 채택
PAY_AMT2		100.0%	✓ 채택
PAY_AMT3		100.0%	✓ 채택
PAY_AMT6		100.0%	✓ 채택
PAY_5		98.8%	✓ 채택
BILL_AMT5		98.8%	✓ 채택
PAY_6		97.5%	✓ 채택
PAY_AMT4		96.2%	✓ 채택
PAY_AMT5		95.0%	✓ 채택
SEX		91.2%	✓ 채택
BILL_AMT6		88.8%	✓ 채택
BILL_AMT2		71.2%	✓ 채택
BILL_AMT3		56.2%	제외
BILL_AMT4		28.7%	제외
EDUCATION		21.2%	제외
MARRIAGE		15.0%	제외

5 학습 설정 + 튜닝 — ensemble

키	값
tuned	NO
params (최종)	{'weights': {'elasticnet': 0.4840579071352653, 'catboost': 0.5159420928647347}}